ReaLLM: Interface-Level Transparency as a Safety Intervention in Human–LLM Interaction

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1. Introduction

The AI safety field has invested heavily in technical interpretability. AI safety research is focused on the transparency and explainability of AI models. Yet, explainable AI (XAI) approaches have been predominantly algorithm-centered, focused on revealing model internals rather than the social context in which AI operates (Ehsan et al., 2021). Mechanistic interpretability reverse-engineers neural circuits; representation engineering reads high-level model representations (Center for AI Safety, 2024). These are important but researcher-facing. Millions of everyday users do not benefit from knowing about polysemantic neurons.

This paper argues that for large language models (LLMs), more accessible layers of social transparency have been neglected, ones written in plain language. System prompts are one such mechanism used by different AI labs. Essentially, guidelines that define model identity, encode prohibitions, and establish guardrails to influence their behavior and responses. But here is the caveat: these instructions, unbeknownst to the user, are attached to their request before reaching the AI model. They are often deliberately hidden from the users. Research has documented a "gulf between user expectation and experience" with conversational agents for nearly a decade (Luger & Sellen, 2016); concealed system prompts explain part of that story.

In a prior article, I analyzed system prompts from major AI labs. This paper builds on that analysis: interface-level transparency is not a replacement for algorithm-centered interpretability, but a complementary mechanism addressing the socio-organizational context that shapes human-AI interaction.

2. What LLM Interfaces Hide

Analysis of system prompts from major AI labs reveals three categories of hidden control:

Identity and Persona. Models are assigned specific personalities. Some adopt a "social chameleon" approach, mirroring user tone; others are configured as "careful and precise" or "deep research assistants."

Behavioral Constraints. System prompts contain layered prohibitions: hard guardrails against weapons information; intellectual guardrails limiting quotations; accuracy directives against hallucination. These operate silently, shaping what users can elicit.

Meta-Guardrails. Most tellingly, models are instructed to conceal their own instructions. OpenAI's prompts forbid disclosure of system messages; Claude is told to withhold reasoning behind refusals "to avoid irritating the user." The system hides behind what it is hiding.

Studies such as Greshake et al. (2023) show that LLM applications "blur the line between data and instructions". Not knowing that a complicated constitution of instructions gets attached to your prompts when you interact with an LLM has direct and indirect consequences. For instance, users hold simplified, often incorrect mental models of LLM ecosystems (Wang et al., 2025).

Indirectly, the lack of understanding exposes users to other risks catalogued by the MIT AI Risk Repository (Slattery et al., 2024): Domain 5.1 identifies "overreliance and unsafe use", users trusting AI without understanding its constructed nature. Domain 7.4 flags opacity as creating mistrust and an inability to identify and correct errors.

The irony is that system prompts are written in natural language and are, in a sense, the most accessible layer of model architecture. On the other hand, it is not so clear as to how these system-level instructions should be communicated to the users and if they could even be all that useful. The next section addresses these tensions.

3. Interface Transparency as a Soft Safety Mechanism

This paper advances a design thesis: selectively exposing hidden layers, the model's given roles, constraint rationales, and uncertainty signals could function as a soft safety intervention, recalibrating user mental models without modifying the underlying model.

This is inference level explainablilty. Technical interpretability asks: *What is the model doing internally?* Interface transparency asks: *What does the user need to know?* The second directly serves the millions making decisions based on AI outputs.

Transparency is not automatically beneficial. Poursabzi-Sangdeh et al. (2021) found interpretable "clear box" models led to *increased* overreliance—transparency created an illusion of understanding. Buçinca et al. (2021) showed explanations did not reduce overreliance; users developed general heuristics rather than evaluating each recommendation.

Yet transparency can work. Vasconcelos et al. (2023) found explanations reduce overreliance when tasks are difficult and disclosure reduces verification costs. The implication is that design matters. Selective, context-sensitive disclosure outperforms both maximal transparency and total opacity.

Trade-offs remain: cognitive load from excessive disclosure; performative transparency creating false confidence; users gaming revealed prompts; the question of who decides what gets revealed. These argue for intentional design, not abandoning transparency.

4. Proof-of-Concept Prototype

To bring these claims and tensions to attention, I developed a minimal prototype. It is a conceptual demonstrator, not an evaluated system.

The prototype implements two features:

1. Users can explicitly view the full system prompt that defines the assistant's role, behavioral constraints, and communicative style.
2. A side panel in the interface that provides brief, context-sensitive annotations that link elements of the model's output to relevant aspects of its role definition or constraints, highlighting why a response takes a particular tone, stance, or level of certainty.

The prototype illustrates how identical outputs can be interpreted differently when contextual layers become visible. A response reading as authoritative in a standard interface becomes provisional when accompanied by uncertainty markers.

5. Conclusion

This paper argues for the necessity of social transparency, and system prompts are a prime candidate. Ehsan et al.'s (2021) 4W framework asks: Who made decisions about the system? What does it do? When? Why? Revealing system prompts answer these questions.

Interface design is a tractable lever. Amershi et al. (2019) synthesized 18 validated guidelines for human-AI interaction, including capability disclosure and confidence communication. The infrastructure exists; what remains is applying it to LLM transparency.

This paper has argued that concealing system prompts is a design choice with safety implications, not a technical necessity. The prototype demonstrates one approach; empirical validation remains future work: we ought to investigate how prompt visibility affects reliance, adaptive disclosure mechanisms, policy questions about standards.

References

Amershi, S., Weld, D., Vorvoreanu, M., Fourney, A., Nushi, B., Collisson, P., Suh, J., Iqbal, S., Bennett, P. N., Inkpen, K., Teevan, J., Kikin-Gil, R., & Horvitz, E. (2019). Guidelines for human-AI interaction. *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*, Article 3, 1–13. https://doi.org/10.1145/3290605.3300233

Buçinca, Z., Malaya, M. B., & Gajos, K. Z. (2021). To trust or to think: Cognitive forcing functions can reduce overreliance on AI in AI-assisted decision-making. *Proceedings of the ACM on Human-Computer Interaction, 5*(CSCW1), Article 188, 1–21. https://doi.org/10.1145/3449287

Center for AI Safety. (2024). *Introduction to AI safety, ethics, and society*. https://www.aisafetybook.com/

Ehsan, U., Liao, Q. V., Muller, M., Riedl, M. O., & Weisz, J. D. (2021). Expanding explainability: Towards social transparency in AI systems. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Article 82, 1–19. https://doi.org/10.1145/3411764.3445188

Greshake, K., Abdelnabi, S., Mishra, S., Endres, C., Holz, T., & Fritz, M. (2023). Not what you've signed up for: Compromising real-world LLM-integrated applications with indirect prompt injection. *Proceedings of the 16th ACM Workshop on Artificial Intelligence and Security*, 79–90. https://doi.org/10.1145/3605764.3623985

Luger, E., & Sellen, A. (2016). "Like having a really bad PA": The gulf between user expectation and experience of conversational agents. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 5286–5297. https://doi.org/10.1145/2858036.2858288

Poursabzi-Sangdeh, F., Goldstein, D. G., Hofman, J. M., Wortman Vaughan, J., & Wallach, H. (2021). Manipulating and measuring model interpretability. *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*, Article 237, 1–52. https://doi.org/10.1145/3411764.3445315

Slattery, P., Saeri, A. K., Grundy, E. A. C., Graham, J., Noetel, M., Uuk, R., Dao, J., Pour, S., Casper, S., & Thompson, N. (2024). *The AI Risk Repository: A comprehensive meta-review, database, and taxonomy of risks from artificial intelligence*. https://doi.org/10.48550/arXiv.2408.12622

Vasconcelos, H., Jörke, M., Grunde-McLaughlin, M., Gerstenberg, T., Bernstein, M. S., & Krishna, R. (2023). Explanations can reduce overreliance on AI systems during decision-making. *Proceedings of the ACM on Human-Computer Interaction, 7*(CSCW1), Article 129, 1–38. https://doi.org/10.1145/3579605

Wang, X., Wang, X., Park, S., & Yao, Y. (2025). Mental models of generative AI chatbot ecosystems. *Proceedings of the 30th International Conference on Intelligent User Interfaces*, 1016–1031. https://doi.org/10.1145/3708359.3712125